probably underlain by more familiar water ice and rock. There are impact basins, which seem to have been flooded with liquid before refreezing (so there once were lakes on Triton); impact craters; long crisscrossing valleys; vast plains covered by freshly fallen nitrogen snow; puckered terrain that resembles the skin of a cantaloupe; and more or less parallel, long, dark streaks that seem to have been blown by the wind, and then deposited on the icy surface—despite how sparse Triton's atmosphere is (about 1/10,000 the thickness of the Earth's).

All the craters on Triton are pristine—as if stamped out by some vast milling device. There are no slumped walls or muted relief. Even with the periodic falling and evaporation of snow, it seems that nothing has eroded the surface of Triton in billions of years. So the craters that were gouged out during the formation of Triton must have all been filled in and covered over by some early global resurfacing event. Triton orbits Neptune in the opposite direction to Neptune's rotation—unlike the situation with the Earth and its moon, and with most of the large moons in the Solar System. If Triton had formed out of the same spinning disk that made Neptune, it ought to be going around Neptune in the same direction that Neptune rotates. So Triton was not made from the original local nebula around Neptune, but arose somewhere else—perhaps far beyond Pluto—and was by chance gravitationally captured when it passed too close to Neptune. This event should have raised enormous solid-body tides in Triton, melting the surface and sweeping away all the past topography.

In some places the surface is as bright and white as freshly fallen Antarctic snows (and may offer a skiing experience unrivaled in all the Solar System). Elsewhere there's a tint, ranging from pink to brown. One possible explanation: Freshly fallen snows of nitrogen, methane, and other hydrocarbons are irradiated by solar ultraviolet light and by electrons trapped in the magnetic field of Neptune, through which Triton plows. We know that such irradiation will convert the snows (like the corresponding gases) to complex, dark, reddish organic sediments, ice tholins—nothing alive, but here too